

Weekly Plan & Progress Report (Week11)

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Week 11 Report (2026-05-08 to 2026-05-14)

1. Milestones Achieved

1. Completed full documentation and structured consolidation of the "AML multi-agent collaboration system (POC)" solution.
 - Deliverables cover business background, input/output definitions, risk grading criteria, evidence chain structure, and final report format.
 - This remains aligned with Week 10's strategy of "converge architecture and interfaces first," with clear implementation boundaries for the next phase.
2. Completed detailed end-to-end interaction design for the three-agent collaboration mechanism.
 - Clarified responsibility boundaries, AgentCard skills, A2A message interactions, and state transitions for HostAgent / MarketAgent / TransactionAgent.
 - Produced a demonstrable sequence flow (intent parsing -> dispatch -> evaluation -> re-dispatch -> report generation).
3. Completed node-level flow design for HostAgent using LangGraph.
 - Solidified the closed-loop logic of `parse_intent`, `plan_investigation`, `dispatch_to_agent`, `evaluate_response`, and `generate_report`.
 - Confirmed an iterative decision mechanism where insufficient evidence triggers further dispatching, supporting multi-round investigations.
4. Completed data-layer and reproducible run-path design.
 - Finalized four SQLite core tables (tasks/messages/artifacts/sessions) and their data lifecycle definitions.
 - Added sample data, initialization flow, CLI real-time display examples, and quick-start guidance to reduce demo setup effort.
5. Completed a standardized output template for the first high-risk case.
 - Using the case "U002 sold a Tesla Model X for \$1,000,000," defined the CRITICAL risk report structure, evidence-chain fields, and recommended action format.

2. Milestones Missed

1. Did not complete a runnable implementation where the three agent services can start independently and call each other.

- Reason: this week still prioritized design/specification solidification, and did not yet enter full coding and integration.
 - Resolution: prioritize the minimum A2A protocol foundation modules next week (task type definitions, server interfaces, client invocation, and service registry), plus the three-agent service skeleton to unblock the minimum closed loop.
2. Did not complete true end-to-end integration and performance validation (including SSE link stability).
- Reason: without executable services and real call logs, stable integration stress testing and failure-recovery validation were not feasible.
 - Resolution: once the minimum loop is running, immediately execute single-case integration, three failure-path regressions (timeout/empty result/call failure), and timing statistics.
3. Did not complete implementation-level validation for reproducible assets (data generation flow + vectorized artifacts + DB initialization run-through).
- Reason: only process definitions and examples are available so far; execution validation and result records are still pending.
 - Resolution: include data generation, vector build, and DB initialization in a one-pass acceptance checklist next week and produce execution records.

3. Plan and Effort Adjustments

- 2026-05-09: adjusted this week's goal from "coding directly" to "completing an executable-level technical specification first."
 - Context: in a multi-agent collaboration scenario, failing to freeze interfaces and state machines first would lead to high integration rework cost.
 - Decision: prioritize unified definitions for protocol, process, data model, and demo path.
- 2026-05-11: split "implementation validation" into a two-phase approach (document validation - > code validation).
 - Context: at this stage, cross-agent responsibility boundaries and evidence-chain interpretation consistency must be ensured first.
 - Decision: deliver a reviewable solution package before moving to minimum runnable implementation.
- 2026-05-13: confirmed Week 12 focus shift to "invoice data structure and persistence logic" for research and implementation preparation.
 - Context: next week's deliverables shifted toward invoice workflow, database table structure, and entity relationship mapping, affecting scope and effort allocation.
 - Decision: make invoice data structure and query-path clarification the Week 12 mainline, while moving the AML multi-agent collaboration solution into parallel "minimum-loop implementation + risk list" closure.

4. Foreseeable Risks and Mitigations

1. Risk: a gap between design and implementation may lengthen next week's integration cycle.
 - Mitigation: use a "minimum closed loop first" strategy (single case, single path, observable logs) and expand incrementally.
2. Risk: A2A protocol implementation and SSE event semantics may mismatch, causing unstable cross-agent calls.
 - Mitigation: freeze unified fields and state enums at the protocol data-structure layer first, then run send/sendSubscribe interoperability tests.
3. Risk: multi-round LLM orchestration may increase latency and cost.
 - Mitigation: enforce call budgets, timeout thresholds, and early-stop conditions; prioritize high-value investigation nodes.
4. Risk: Week 12 theme shift (invoice DB structure and relationship clarification) runs in parallel with current AML multi-agent closure, creating resource contention.
 - Mitigation: split work into primary/secondary tracks: primary track on invoice workflow, table structure, and mapping relationships; secondary track only on minimum-loop AML closure and key risk convergence.

Week 12 Plan (2026-05-15 to 2026-05-21)

1. Work Allocation

1. Clarify invoice business flow and scope requirements (primary track)
 - Align with the confirmed meeting flow: enter invoice module, select supplier, input quantity and unit price, calculate tax, create invoice, and verify via lookup.
 - Deliverables: invoice creation process notes, key business field checklist, and exception scenario list (e.g., validation errors during creation).
2. Clarify Postgres database structure and produce a data dictionary (primary track)
 - Organize permission/business-related DBs/schemas (e.g., Authority, Public) and core tables: company, entity, project, and invoice.
 - Deliverables: DB structure diagram, core table field dictionary, and primary/foreign key relationship notes.
3. Validate invoice relationships and query path (primary track)

- Focus on validating the relationship path "company UUID -> invoice list -> invoice-project mapping -> project details," covering buyer/supplier relationships and multi-project invoice scenarios.
 - Deliverables: standard query path guide, representative samples (including recalled status and multi-project mapping cases), and validation records.
4. Close the minimum loop for the AML multi-agent collaboration solution (secondary track)
- Complete minimum three-agent call-chain validation with one runnable case and collect issue backlog.
 - Deliverables: integration records, known issues, and prioritized fix recommendations.

2. Expected Milestones

1. Complete scope freeze for invoice business flow, key fields, and exception scenarios.
2. Complete documentation for core DB table structure, field dictionary, and entity relationships.
3. Complete query-path validation from "company UUID to invoice and project mapping" with sample records.
4. Complete minimum-loop integration validation for the AML multi-agent collaboration solution and produce a risk/issue list.
5. Produce reusable Week 12 deliverables (process notes, query checklist, and acceptance criteria).